

Script-Controlled ACL – Restrict Record Access Based on Field Value

Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.

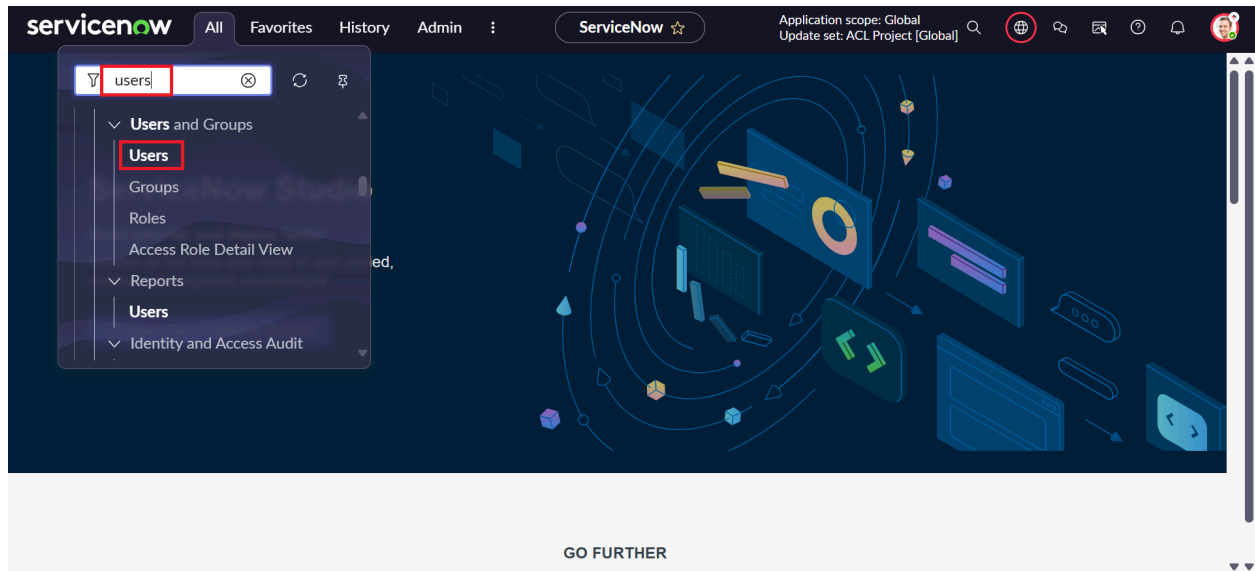
Timer :

3 hrs

Milestone-1 Creation of Users and Roles

User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
 - Navigate to **User Administration → Users**
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record



User ID:
 First name:
 Last name:
 Title:
 Department:
 Location:

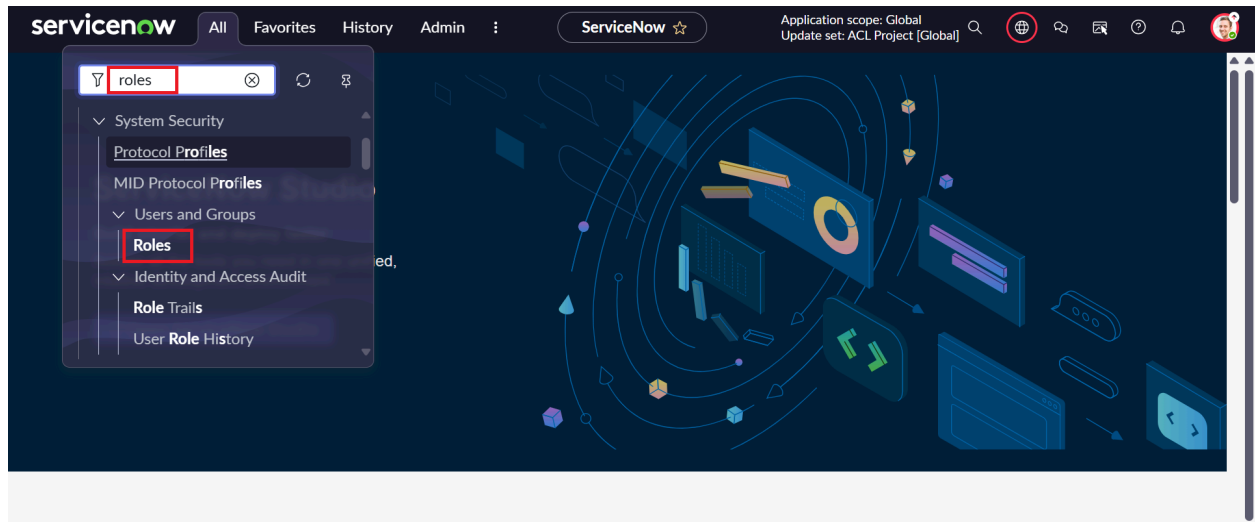
Password needs reset: ☐
 Locked out: ☐
 Active: ☒
 Internal Integration User: ☐

Email:
 Identity type:
 Language:
 Calendar integration:
 Time zone:
 Date format:
 Business phone:
 Mobile phone:
 Photo: [Click to add...](#)

Update Set Password Delete

Role Creation

- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as bb1
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User



Role BB1

Name BB1 Application Global

Elevated privilege ☐

Description

Update Delete

[SN Utils\ Versions \(1\)](#)

Tables Creation

1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false

The screenshot shows the ServiceNow user interface. At the top, the 'Tables' menu is open, with 'Tables' highlighted under the 'System Definition' section. Below this, the 'Table institution Details' configuration page is visible. The page includes a header with navigation links and a toolbar with 'Delete', 'Update', and 'Delete All Records' buttons. A descriptive text box explains that a table is a collection of records in the database. The main configuration area contains two fields: '* Label' with the value 'Institution Details' and '* Name' with the value 'u_institution_details'. To the right, there are fields for 'Application' (set to 'Global') and 'Remote Table' (set to 'Global').

3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
 - **Student Roll Number**– Auto Number
 - **Student Name** – Reference- User
 - **Faculty Name** – Reference- User
 - **Branch** – Choice (ECE, EEE, CSE)
 - **Email**– String
 - **Phone Number**– String
 - **Description**– Multi String
5. Save the Table

Table Columns for text Search 1 to 13 of 13

Dictionary Entries

Column label	Type	Reference	Max length	Default value
Updates	Integer	(empty)	40	
Created by	String	(empty)	40	
Student Roll Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();
Description	String	(empty)	1,000	
Updated by	String	(empty)	40	
Phone Number	String	(empty)	40	
Sys ID	Sys ID (GUID)	(empty)	32	
Created	Date/Time	(empty)	40	
Faculty name	Reference	User	32	
Branch	String	(empty)	40	
Updated	Date/Time	(empty)	40	
Student Name	Reference	User	32	
Email	String	(empty)	40	
Insert a new row...				

- Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

institution Details Student Roll Numbe Search Actions on selected rows...

All

Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

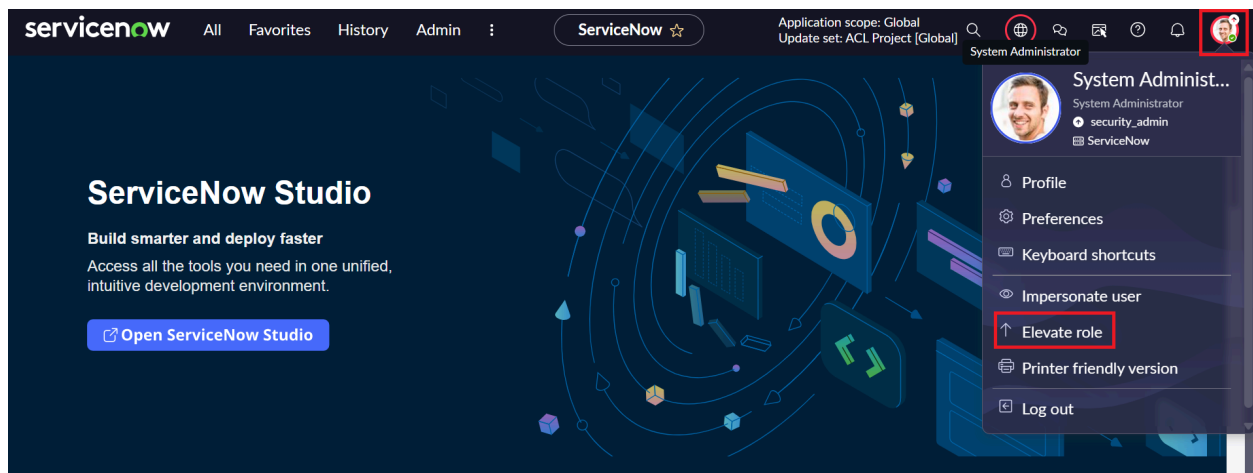
Creation of Access Control Lists(Read ACL)

- In the **Application Navigator**, search for **System Security** → **Access Control (ACL)**.
- Elevate “Security_admin” role
- Click **New** to create a new ACL.
- Enter the following details:
 - Type:** record
 - Operation:** read
 - Name:** u_institution_details
 - Active:** true

- **Advanced:** true
- 5. In the **Requires role** related list, add:
 - bb1 (custom role)
- 6. Add Data condition as **Branch is EEE**
- 7. In the **Script** field, add the following script:

```
(function () {  
  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    // Deny access for all others  
    return false;  
})();
```

- 8. Click **Submit** to save the ACL.



Elevate role

 Close dialog

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

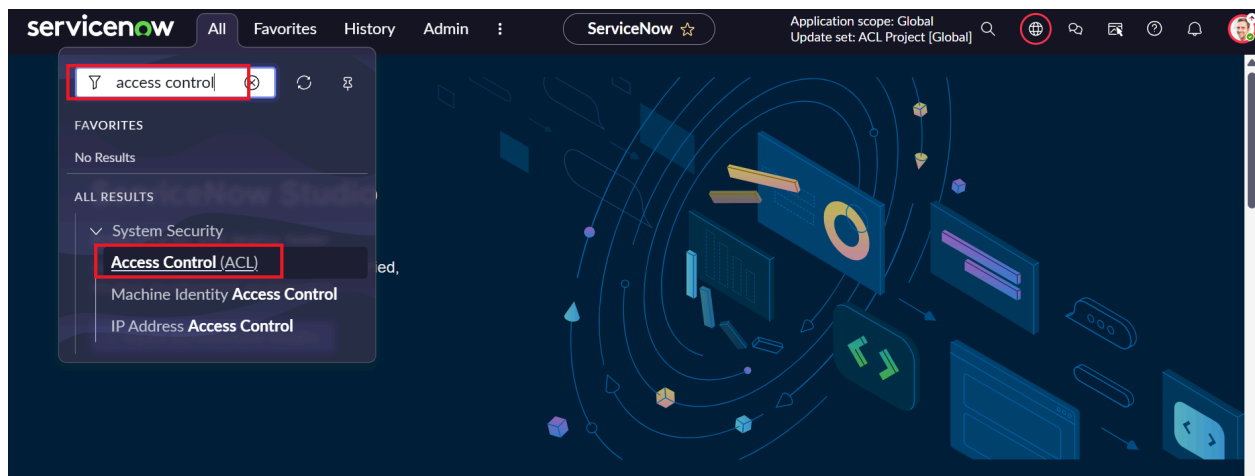
AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel

Update



9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

Verify :

- User with **bb1** role can view **only EEE branch records**

institution Details Student Roll Numbe Search ⚙️ 📄 Actions on selected rows...						
All						
☐ 🔍 Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

- User without the role cannot view any records

institution Details Student Roll Numbe Search ⚙️ 📄						
☐ 🔍 Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
Search	Search	Search	Search	Search	Search	Search



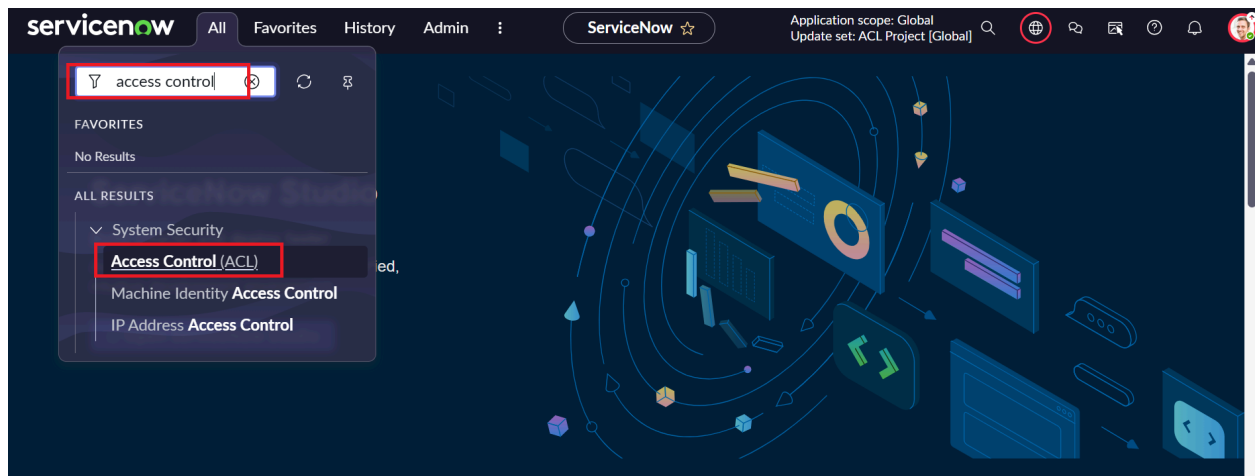
No records to display

- Admin user can view **all records regardless of branch**

institution Details Student Roll Numbe Search ⚙️ 📄 Actions on selected rows... New						
All						
☐ 🔍 Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number	
Search	Search	Search	Search	Search	Search	
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906	
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989	
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090	
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765	
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989	
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090	

Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



Access Controls

Updated

Search

Actions on selected rows...

New

All

Q

Name

Decision Type

Operation

Type

Active

Updated by

Updated ▾

Search

Search

Search

Search

Search

Search

Search

sn_try_build_agent_message

Deny Unless

read

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_freemium_usage_log

Deny Unless

read

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_conversation

Deny Unless

delete

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_conversation

Deny Unless

write

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_message

Deny Unless

delete

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_event_telemetry

Deny Unless

report_view

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_conversation

Deny Unless

report_view

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_task_telemetry

Deny Unless

write

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_task_telemetry

Deny Unless

read

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_event_telemetry

Deny Unless

delete

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_task_telemetry

Allow If

read

record

true

system

about 3 months ago

sn_try_build_agent_knowledge_source

Allow If

write

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_event_telemetry

Allow If

report_view

record

true

system

19-10-2025 22:23:04

①

61

to 80 of 32,702

Access Control u_institution_details

* Type: record

* Operation: create

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Application: Global

Active: ☒

Advanced: ☐

Update Delete

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
BB2
Insert a new row...

Security Attribute Condition

Verify :

- User with **bb1 & bb2** roles can view **only EEE branch records** b New Button

institution Details Student Roll Numbe Search

Actions on selected rows... New

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

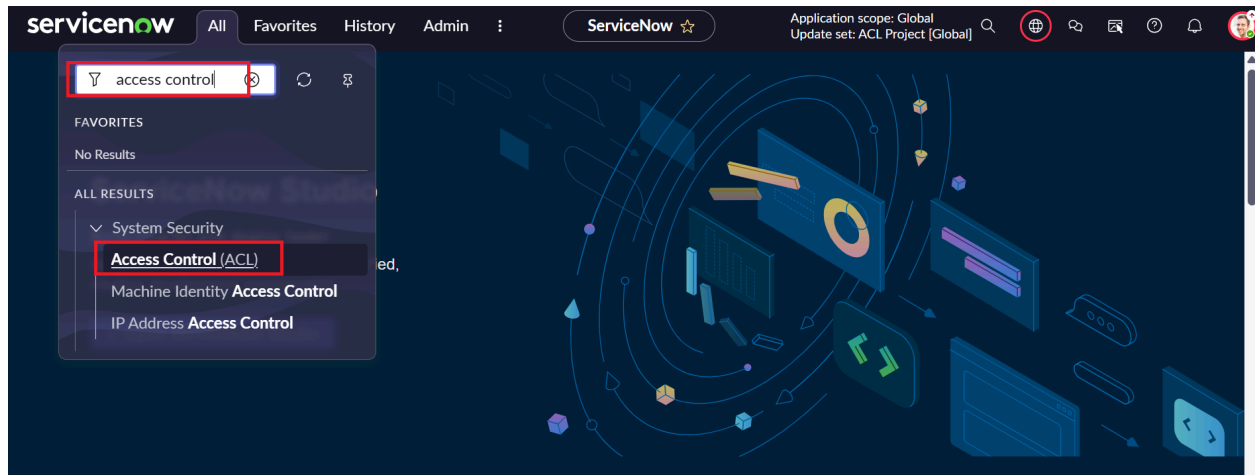
Creation of Access Control Lists(Write ACL)

12. Click New to create a new ACL(Write)

13. Enter the following details:

- Type: record
- Operation: Write
- Name: u_institution_details

- Active: true
14. In the Requires role related list, add:
- bb3 (custom role)
15. No need to add any Data condition
16. Save the ACL Record



Access Controls

Updated

Search

Actions on selected rows...

New

All

☐		Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐								
		sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago
☐		sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04

61

to 80 of 32,702

Access Control

u_institution_details

Update

Delete

* Type

record

* Operation

write

Decision Type

Allow If

Application

Global

Active

☒

Advanced

☐

Admin overrides

☒

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

-- None --

Description

Applies To

No. of records matching the condition: 6

Add Filter Condition

Add OR Clause

-- choose field --

-- oper --

-- value --

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

× BB3

+ Insert a new row...

Verify :

- User with **bb1 &bb2 &bb3** roles can view and edit the **EEE branch records along with New Button**

institution Details							
Student Roll Number		Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007		EEE		com	Abel Tuter	9090898765	Guna Sai
INS0001004		EEE		il.com	Abel Tuter	9090909090	Prashanth Reddy

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
	INS0001007	EEE	ACL Is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
	INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Creation of Access Control Lists(Delete ACL)

17. Click **New** to create a new ACL(Delete)

18. Enter the following details:

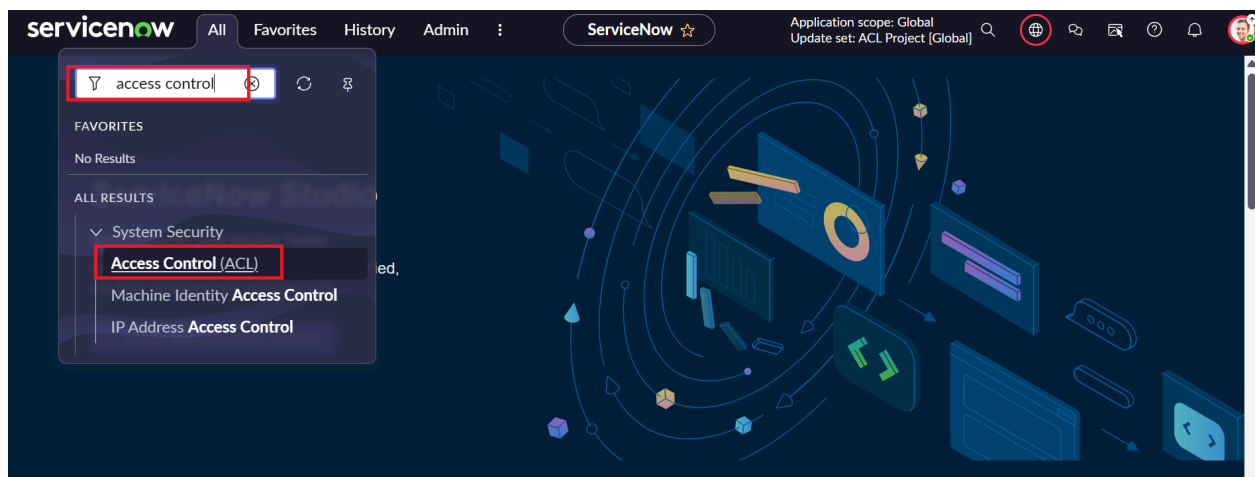
- **Type:** record
- **Operation:** Delete
- **Name:** u_institution_details
- **Active:** true

19. In the **Requires role** related list, add:

- bb4 (custom role)

20. No need to add any Data condition

21. Save the ACL Record



Access Controls Updated Search							
All							
Name	Decision Type	Operation	Type	Active	Updated by	Updated	
Search	Search	Search	Search	Search	Search	Search	Search
sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago	
sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04	

Access Control

New record

Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type

record

* Operation

delete

Decision Type

Allow If

Application

Global

Active

☒

Advanced

☐

Admin overrides

☒

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

-- None --

Description

Applies To

No. of records matching the condition: 6

Add Filter Condition

Add OR Clause

-- choose field --

-- oper --

-- value --

Requires role

Role

BB4

Insert a new row...

Security Attribute Condition

Verify :

- Users with **bb1 &bb2 &bb3&bb4** roles can see, create and edit along with delete the **EEE branch records**.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
☐	INS0001007	EEE	ACL is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
☐	INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Num

INS0001007

EEE

ACL is Executed

saiguna@gmail.com

Abel Tuter

90908987

INS0001004

EEE

ACL is Executed

prashanth@gmail.com

Abel Tuter

90909090

Actions on selected rows...

Delete

Delete with preview...

Create Application File

Assign Tag:

New tag

Now Platform Security

Strategic Portfolio Management

Field Service Management

Now Platform

Security

IT Asset Management

IT Service Management

IT Operations Management

More...

institution Details

Student Roll Numbe

Search

Actions on selected rows...

All

Student Roll Number

Branch

Description

Faculty name

Phone Number

Student Name

INS0001007

EEE

ACL Is Executed

Abel Tuter

9090898765

Guna Sai

INS0001004

EEE

ACL Is Executed

Abel Tuter

9090909090

Prashanth Reddy

Confirmation

Delete the selected record(s)?

Cancel

Delete

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.